

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch: Common to all branches

Semester: II

Subject Code & Name: (BTBS201) Engineering Mathematics - II

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The value of $\sin i y = \dots$	Remember (CO1)	1
	a. $\sinh y$ b. $i \sinh y$ c. $i \sin y$ d. $-i \sin y$		
2	The imaginary part of $(\cos 5x + i \sin 5x)^3 =$	Remember (CO1)	1
	a. $\cos 3x$ b. $\sin 3x$ c. $\cos 15x$ d. $\sin 15x$		
3	If z_1 and z_2 are two complex numbers then amplitude of $z_1 z_2$ is	Remember (CO1)	1
	a. Quotient of their amplitudes b. Product of their amplitudes c. Addition of their amplitudes d. Subtraction of their amplitudes		
4	The differential equation $(x + x^8 + ay^2)dx + (y^8 - y + bxy)dy = 0$ is exact if	Understand (CO2)	1
	a. $b \in 2a$ b. $b = a$ c. $a \neq b$ d. None		
5	To obtain the orthogonal trajectory of $f(x, y, c) = 0$, $\frac{dy}{dx}$ is replaced by	Remember (CO2)	1
	a. $\frac{dy}{dx}$ b. $\frac{dx}{dy}$ c. $-\frac{dx}{dy}$ d. $\frac{dr}{d\theta}$		
6	The order of the differential equation $\frac{dy}{dx} + 3xy = \tan x$ is	Understand (CO2)	1
	a. 3 b. 2 c. 1 d. 4		
7	The particular integral (P. I.) of $(D^2 + 3)y = 3$ is	Understand (CO3)	1
	a. 3 b. 1 c. 2 d. 6		
8	The complementary function of the linear differential equation $f(D)y = X$ having distinct real roots m_1 and m_2 of the auxiliary equation is	Remember (CO3)	1
	a. $y = c_1 e^{-m_1 x} + c_2 e^{-m_2 x}$ b. $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$		
	c. $y = c_1 e^{m_1 x}$ d. $y = c_2 e^{m_2 x}$		
9	The half range Fourier Cosine series for $f(x)$ defined in the interval $0 \leq x \leq l$ is	Remember (CO4)	1
	a. $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$ b. $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$		

	c. $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$		d. $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \sin \frac{n\pi x}{l}$			
10	The constant a_0 in Fourier series expansion of function $f(x) = x$ in $(-\pi, \pi)$ is given by				Understand (CO4)	1
	a. π	b. 0	c. $\frac{\pi^2}{12}$	d. $\frac{\pi^2}{6}$		
11	If $\vec{r} = xi + yj + zk$ then $\nabla \times \vec{r} =$				Understand (CO5)	1
	a. 0	b. 3	c. 2	d. $\vec{r}$		
12	$\nabla \cdot \vec{F}$ is				Remember (CO5)	1
	a. Scalar quantity	b. Vector quantity	c. Unit vector	d. None of these		
Q. 2	Solve the following.					12
A)	Show that $(1+i)^n + (1-i)^n = 2^{\frac{n+2}{2}} \cos \frac{n\pi}{4}$				Understand (CO1)	6
B)	Show that $\cos \left[i \log \left(\frac{a-ib}{a+ib} \right) \right] = \frac{a^2-b^2}{a^2+b^2}$				Understand (CO1)	6
Q.3	Solve the following.					12
A)	Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$				Apply (CO2)	6
B)	Find the orthogonal trajectories of the curves $x^2 + 2y^2 = c^2$				Remember (CO2)	6
Q. 4	Solve Any Two of the following.					12
A)	Solve by variation of parameters $(D^2 + 1)y = \sec x$				Apply (CO3)	6
B)	Solve $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + 4y = \cos(\log x) + \sin(\log x)$				Apply (CO3)	6
C)	Solve $(D^2 - 2D + 1)y = xe^x \sin x$				Apply (CO3)	6
Q.5	Solve Any Two of the following.					12
A)	Find the Fourier series of $f(x) = \frac{1}{2}(\pi - x)$ with period of (2π) to be valid in the interval $(0 \leq x \leq 2\pi)$.				Remember (CO4)	6
B)	Find the Fourier series for the function $f(x) = x $ in $(-1, 1)$.				Remember (CO4)	6
C)	Find half range cosine series for the function $f(x) = x^2 - 2$ in the interval $0 < x < 2$.				Remember (CO4)	6
Q. 6	Solve Any Two of the following.					12
A)	Find grad ϕ at $(1, 1, 1)$ if $\phi = x^2y + xy^2 + yz^2$ and find $\text{div} \vec{A}$ if $\vec{A} = 2x^2z \vec{i} - xyz \vec{j} + 3yz^2 \vec{k}$.				Remember (CO5)	6
B)	If $\vec{r} = xi + yj + zk$ then show that $\text{div}(r^n \vec{r}) = (n+3)r^n$.				Understand (CO5)	6
C)	A resistance of 100Ω , an inductance of $0.5H$ is connected in series with a battery of $20V$. Find the current in circuit as a function of t .				Remember (CO2)	6
*** End ***						